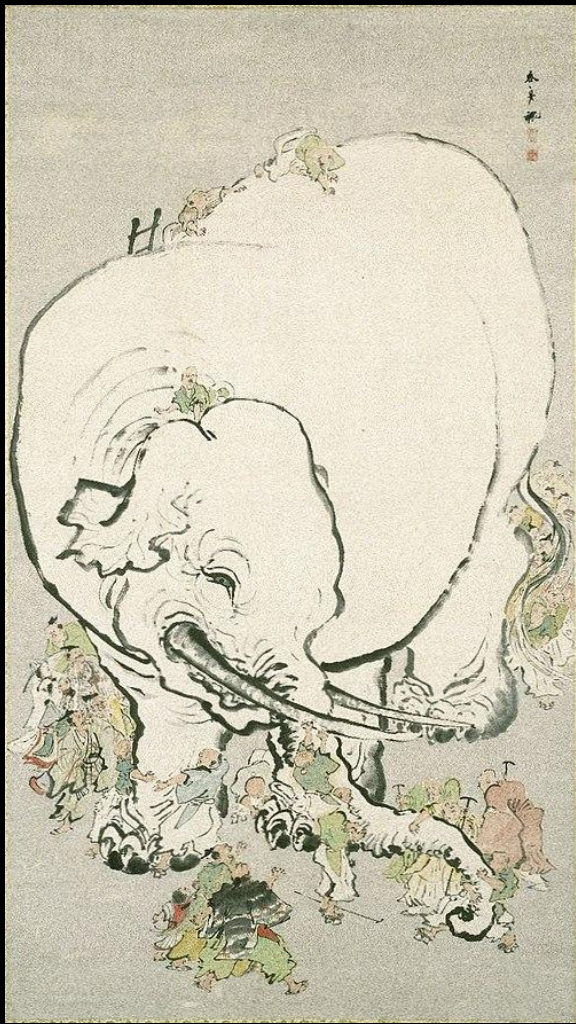




Part III:

The blind men and the elephant

Three aims:

1. Understand **ethical issues** that apply to the use of genAI
2. Explore how ethical issues can be **addressed**
3. Draft **strategies** for addressing ethical issues

1. Ethics.

Duty/virtue ethics vs teleological

Question: is it OK to kill someone?

Duty/virtue ethics vs teleological

Question: is it OK to kill someone?

What if that person was Hitler in 1938?

What if that person was going to kill someone?

What if that person was going to kill 10 people?

What if that person was going to kill you?

Competing ethics

Is it OK to kill someone?

Is it OK to let someone be killed?

Competing rights

Is it OK to reveal private information about a person?

Right to privacy vs accountability

Is it OK to reveal private information about a person?

If their salary is paid from public (your) money?

If that contradicts their public position?

If it exposes unethical behaviour?

It it exposes illegal behaviour?

“Parkinson's UK [used machine learning] and **increased revenue** in its appeal by over £15,000. It also identified reductions in its list size, therefore **bringing costs down.**”

It's not the law:

- Different **ethical models**, e.g. duty vs consequentialist
- Considering different **ethical tensions**, e.g. privacy versus public interest
- Environmental harm vs saving lives
- Environmental harm vs environmental protection
- Labour exploitation vs job creation

A SWOT analysis

Outline SWOT* for AI usage in one of the following scenarios, focusing on **accuracy, bias & employment**:

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
- Analyse data on user behaviour

*SWOT = Strengths Weaknesses
Opportunities & Threats

Then assess those risks

Use the [risk assessment template](#) to assess the **risks** around those areas for [your scenario](#)

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
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2. Issues.

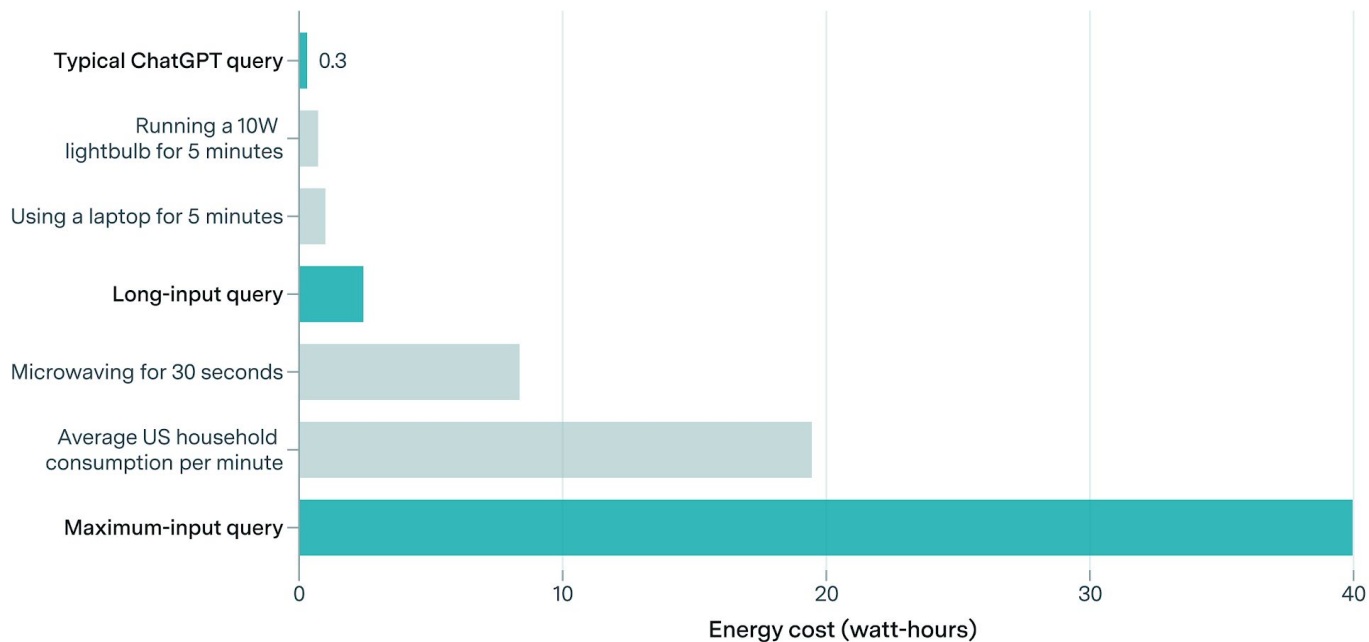
Environment.

Questions identified

1. What is the impact of using AI?
2. Which types of AI are more energy intensive?
3. How can we quantify the impact of our AI use?
4. How can we protect the environment/mitigate impact?
5. How do you navigate concerns effectively?

[Answers compiled in this post.](#)

Energy consumption per ChatGPT query is small compared to everyday electricity use



Pessimistic estimates of the energy usage of ChatGPT with GPT-4o across for different query lengths: typical (<100 words), long (~7,500 words), and maximum context length (~75,000 words), with an average response length of 400 words.

57-3,353 joules (smallest-largest model)

“or an estimated 114-6,706 joules when accounting for cooling, other computations, and other demands. This is about what it takes to ride 6-400 feet on an e-bike, or run a microwave for one-tenth of a second to 8 seconds.”

“In the early 2010s, data centers themselves faced dire predictions. [Then blockchain and cryptocurrency]. In reality, those worst-case scenarios didn’t materialize – thanks to huge improvements in efficiency [and use of renewables]”

...But that's just typical queries

1. Reasoning models “think” for longer and use more energy
2. Bigger models use more energy
3. Images often use more energy — but not always
4. And video...

x42,000 for 5-sec video

“Enough to power a microwave for over an hour”

Strategies

- Turning off automated AI [functionality](#)
- Switching to AI-opt-out alternatives (e.g. [DuckDuckGo](#))
- Comparing AI vs non-AI approaches*, or [different models](#)
- More efficient interactions: plan prompts & templates
- Measuring impact for accountability/monitoring
- (Please) **stop being polite** (sorry)

“We realised there was an inherent contradiction in the fact that we would be reporting about the climate crisis, but taking loads of flights to do so. We decided to take it as an opportunity to assess the carbon emissions of our own reporting.”

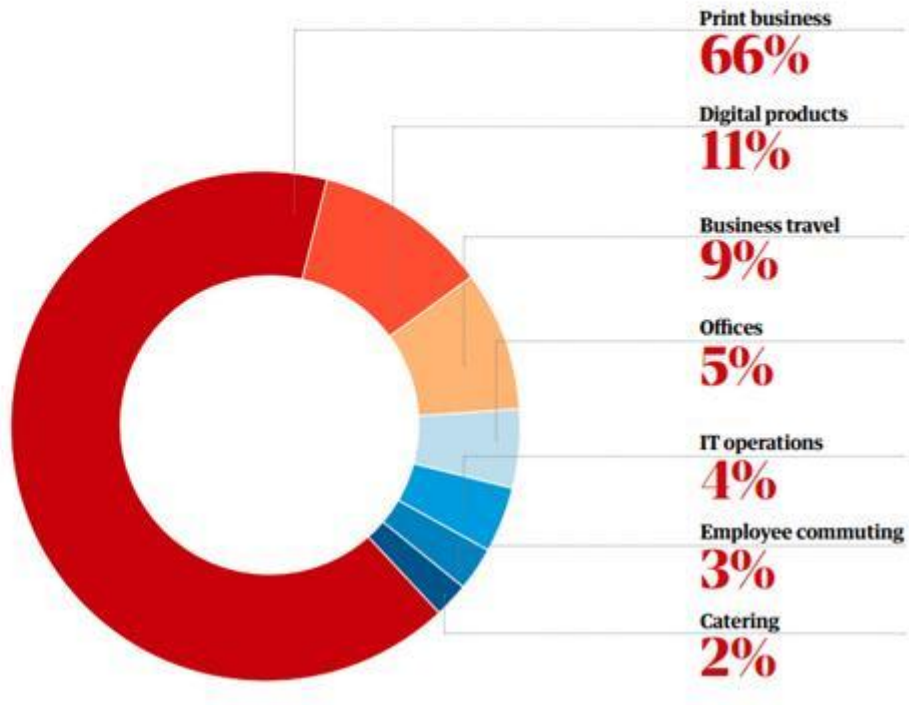
Where our greenhouse gas emissions come from

Greenhouse gas emissions 2023/24, tCO₂e

Our print business remains the biggest source of emissions but has fallen as a proportion of total emissions from 73% in 2019/20 to 66% in 2023/24.

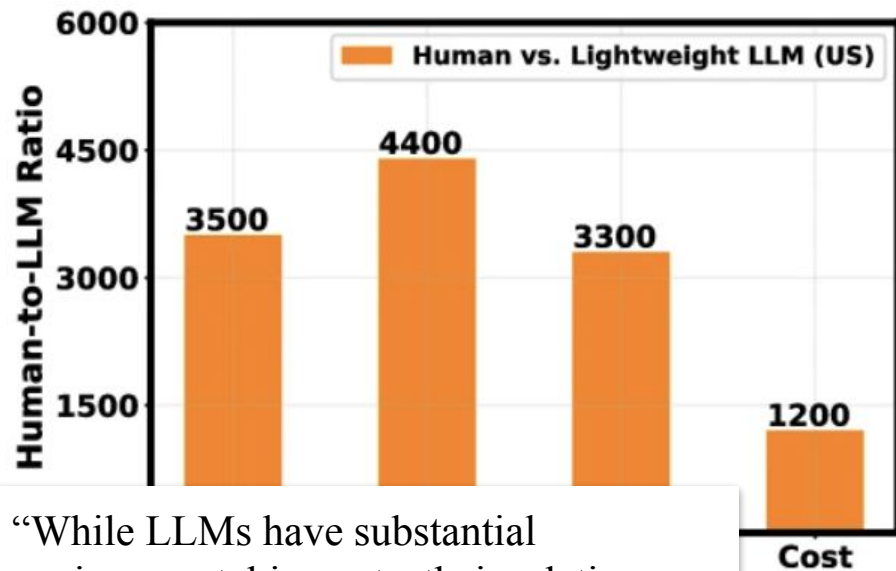


“We worked with an external environmental consultancy to map out all the different emissions in our operations and our supply chain. “Getting that initial data helped us to establish the total size and sources of our emissions, and set targets that we thought were achievable.”

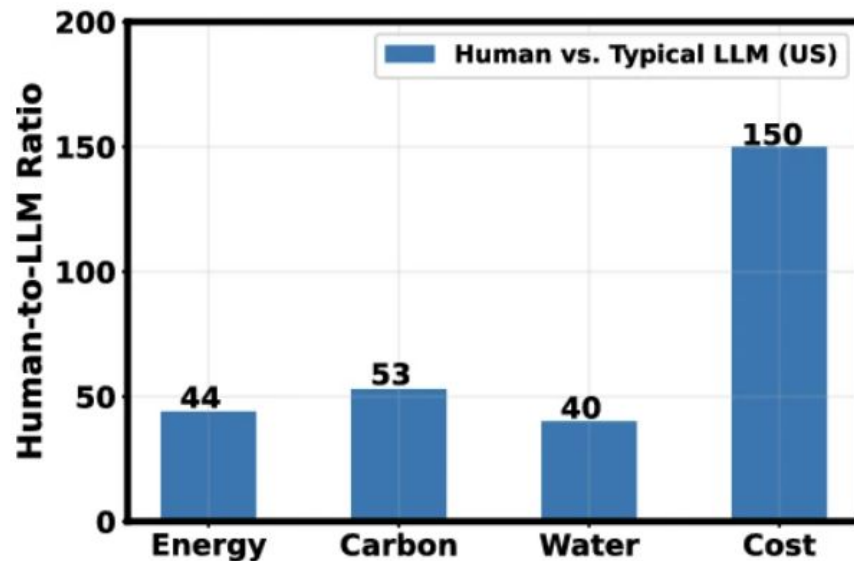


[Julie Richards, Director of Sustainability and Operational Transformation, The Guardian](#)

“When a company employs someone, they are not just paying for task-specific energy expenditure. They are effectively renting a portion of that person’s life — typically 40 hours per week. During this time, all of the individual’s environmental impacts, from commuting to basic life functions, are part of the cost of their labor.”



“While LLMs have substantial environmental impacts, their relative impacts can be dramatically lower than human labor in the U.S. for the same output, with human-to-LLM ratios ranging from 40 to 150 for a typical LLM (Llama-3-70B) and from 1200 to 4400 for a lightweight LLM (Gemma-2B-it).”



Reconciling the contrasting
narratives on the environmental
impact of large language models |
Scientific Reports

Prompt design techniques

More resource intensive:

- **Recursive prompting, CoT (Chain of Thought)**

Less resource intensive:

- **Negative prompting**

Both (individually more, overall potentially less):

- **RAG (Retrieval Augmented Generation)**
- **Meta-prompting**

Update the risk assessment

Assess the **environmental risks** of using genAI for your scenario:

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
- Analyse data on user behaviour

Power.

“Misplaced focus on individual use of technology can **divert us from the bigger picture and real solutions**. Big Oil applied a similar tactic in the climate arena. BP popularized the concept of the “carbon footprint” calculator, implicitly telling us that climate change is due to our individual choices”

“After reporting on this for months, I think there are **more important questions.**

“Water being drained from aquifers in the country’s driest state to power data centers that are drawn to the area by **tax incentives and easy permitting processes.** Or how Meta’s largest data center project is relying on **natural gas despite industry promises to use clean energy.** Or the fact that nuclear energy is not the silver bullet that AI companies often make it out to be.”

Questions raised about power

- How do we hold to account for problems caused by AI?
- How does genAI concentrate power?
- Where can AI be used to empower (e.g. reasonable adjustments to documents)?
- How can it improve access to justice?
- How might it affect the quality of legal advice?
- How do we ensure [artists] are properly compensated?
- What regulation do we need?

Issues

- Lack of transparency about training data and energy use
- [Labour](#) practices (content moderation and labelling)
- Spread of AI-generated “slop” and misinformation
- Perpetuation of biases and inequality (e.g. digital divide)
- Lobbying to weaken [laws](#)
- [Responsible AI](#) & human rights [policies](#) lack detail
- Our/clients’ power and accountability

Strategies?

- **Policies and guidelines:** about the ethical debates, risk assessments, strategies, funding choices etc.
- **Comparing** models, e.g. [Comma](#) (©), local LLMs
- Licensing: check/negotiate **contracts**
- **Transparency:** about the use of AI in work (threshold?)
- **Monitoring:** labour, implementation of AI policies, enforcement of laws
- **Countering, campaigning & empowering:** slop/misinformation, biases/inequalities

Update the SWOT

Assess the opportunities and threats around using genAI for your scenario around **power & accountability** :

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
- Analyse data on user behaviour

Security.

Data protection/privacy

1. Who has access to the information entered?
2. What limits should we set on data that can be entered?
3. How do we approach genAI in recruitment?
4. What links are there between genAI and cyber security threats?

“We will not use your Inputs or Outputs to train our models, unless: (1) your conversations are **flagged for Trust & Safety review** ... (2) you’ve explicitly **reported** the materials to us (for example via our feedback mechanisms), or (3) you've otherwise **explicitly opted in** to the use of your Inputs and Outputs for training purposes.”

“Gemini Apps Activity is on by default if you are 18 or older ...

To stop future conversations from being reviewed or used to improve Google machine-learning technologies, turn off Gemini Apps Activity. You can also review and delete past conversations in Gemini Apps Activity.”

Settings



 General

Improve the model for everyone

On >

 Personalization

Shared links

Manage

 Speech

 Data controls

Export data

Export

 Security

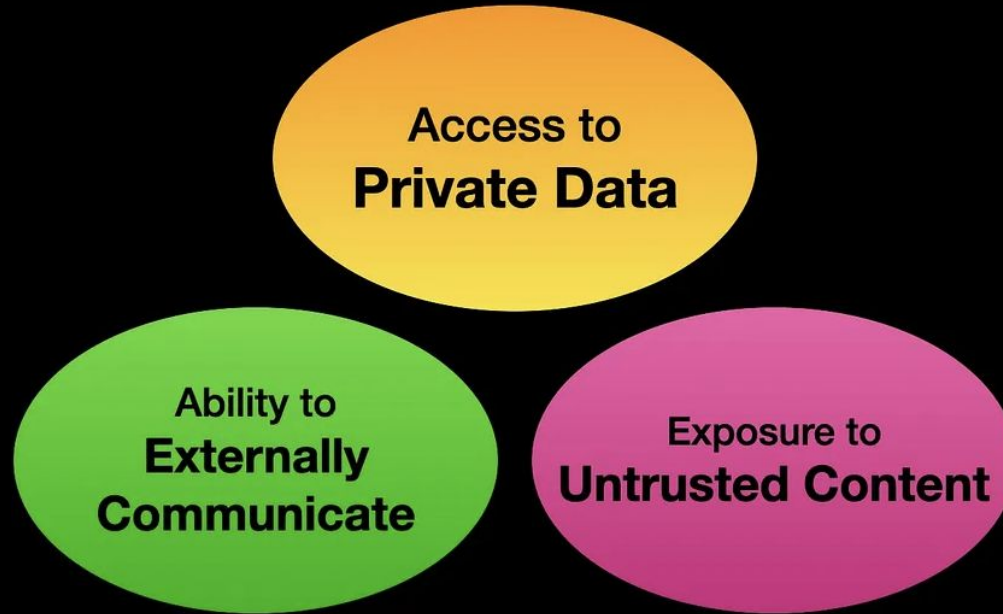
Delete account

Delete

Risks

- Memorisation and association (with a user)
- Training data extraction/membership inference attacks
- 'Snitching' to authorities/media
- New “attack vectors”, e.g. website chatbot, internal LLM
- AI agents...

The lethal trifecta



“If your agent combines these three features, an attacker can **easily trick** it into accessing your private data and sending it to that attacker.”

Strategies before using AI

- **Opting out** of data-sharing (including [outside AI](#))
- **Review [privacy policy](#)/switch** to [enterprise](#) versions
- Blocking AI scrapers in **robots.txt** on your website
- **Comparing** AI vs non-AI approaches, or different models
- [Update] **policies/guidelines**: when/where AI can't be used, **HITL** (human-in-the-loop) + [access control](#)
- **Data minimisation, consent and purpose limitation**

“Parkinsons UK [uses] its own version of ChatGPT: Pbot. ‘It gives us the full benefit of OpenAI’s technology but **keeps all our sensitive information within our own system to protect confidentiality and comply with data protection regulations.**’”

Strategies with AI

- **Anonymisation & de-identification** (esp RAG)
- **Prompt templating** to embed best practice
- **Local** usage (downloading the LLM)
- **Incident response plan:** [detection, notification, mitigation/remediation, communication, legal](#)
- **AI taskforce** group to review and update all the above!
- The NIST AI Risk Management Framework ([AI RMF](#))

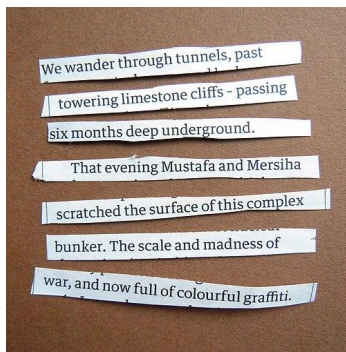
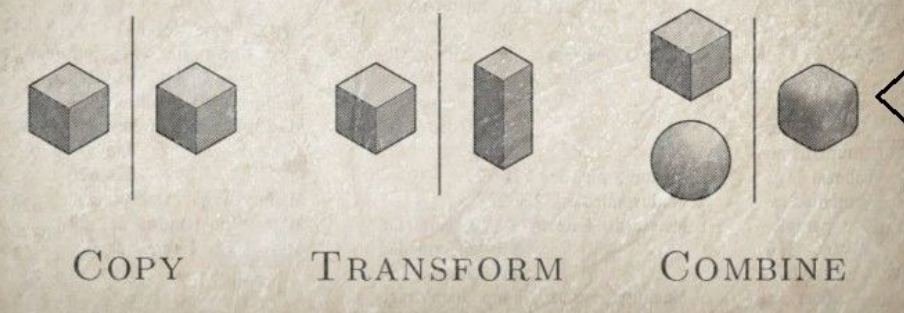
Creativity.

Questions about creativity

1. How do we protect creativity? [From what?]
2. How does AI affect creativity?
3. What is creativity? (Romantic vs rational)
4. How can AI help us think in new/different ways?
5. How can AI change/improve written content?
6. Why is AI more problematic for creatives than [doctors](#)?
7. What does it mean to be human now? The jagged edge?

A **creative** artifact (i.e., a unit of culture: an idea, or process, or product) is one that a consensus of people in a field finds **`novel and appropriate'** (i.e., **`new and useful and surprising'**), and also **solves a problem** (or, problems).

THE BASIC ELEMENTS OF CREATIVITY



The SCAMPER model

Seven perspectives to provoke creative solutions to challenging problems.

S	C	A	M	P	E	R
Substitute	Combine	Adapt	Modify	Put to another use	Eliminate	Reverse
Replace a part, material, or process with something else.	Join elements, ideas, or functions together in new ways – or find a new element you can merge with.	Modify something to better suit a new purpose, person or context.	Enlarge, reduce, change the shape, or alter attributes. Can a small change have a big effect?	Rather than changing the thing itself, consider changing the context it exists in.	Remove elements, simplify, or pare down to essentials. Is less more?	Flip the script, re-order your priorities, invert cause and effect, and turn it all upside-down.

<https://storyality.wordpress.com/2012/12/06/what-is-creativity-and-how-does-it-work>
<https://www.bitesizelearning.co.uk/resources/scamper-model-creativity>

“Oh, there’s no question [computers are creative]. Yes, yes, a million times yes.
Creativity is simple; consciousness, intelligence, those are hard.”

Musician and composer Prof. David Cope



<https://computerhistory.org/blog/algorithmic-music-david-cope-and-emi/>

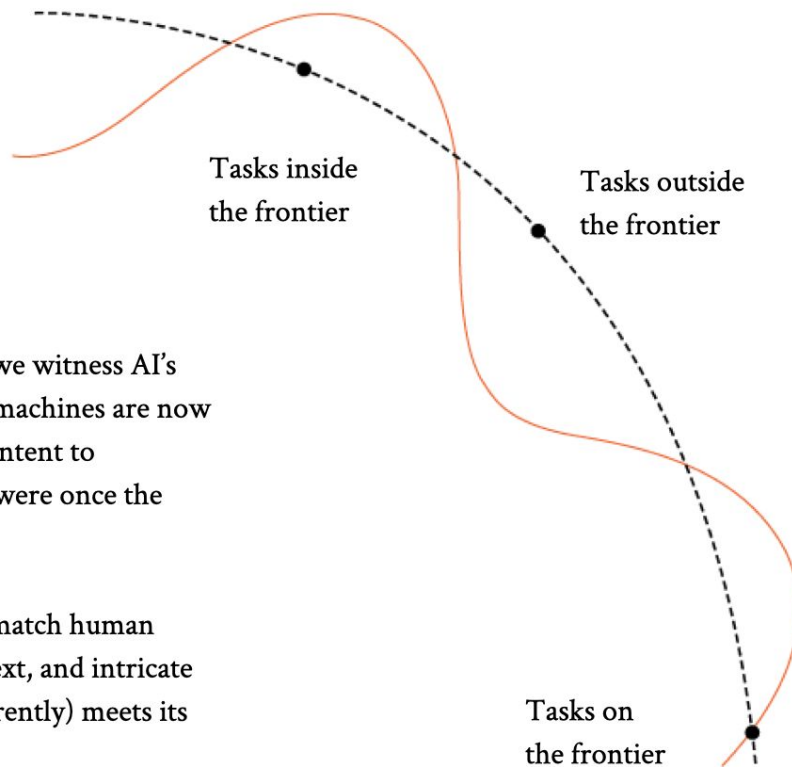
"Like any reader aspiring to be a writer, Anthropic's LLMs trained upon works not to race ahead and replicate or supplant them — but to turn a hard corner and create something different."

Judge William Alsup, Anthropic copyright case

THE JAGGED FRONTIER

The jagged frontier works as follows: on one end of the spectrum, we witness AI's remarkable prowess — tasks that once seemed insurmountable for machines are now executed with precision and efficiency (from generating creative content to predicting complex patterns), with AI showcasing capabilities that were once the exclusive domain of human cognition.

Yet, on the flip side, there are tasks where AI falters, struggling to match human intuition and adaptability. These are areas marked by nuance, context, and intricate decision-making — realms where the binary logic of machines (currently) meets its match.



'Stewardship'

- **Translating** intentions into prompts (“scaffolding”)
- **Steering** (iteration/recursive prompting, RAG)
- **Evaluating** (inc. establishing criteria for that)
- **Human-in-the-loop (HITL)**: checks and verification
- **Selection, editing and adapting** (for your purposes/audience/code)

Table 5.1: Sawyer's Eight Stages of the Creative Process, and How They Correspond to Other Process Models

	Wallas (1926)	Creative Problem Solving (Isaksen, Dorval, & Treffinger, 2000)	IDEAL cycle (Bransford & Stein, 1984)	Robert Sternberg (2006)	Possibility thinking (Burnard, Craft, & Grainger, 2006)	UK QCA (QCA, 2005)	Synectics (Gordon, 1961)	Mumford's group (Scott et al., 2004)	IDEO (Kelley, 2001)
Find the problem		Framing problems	Identify problems, define goals	Redefine problems	Posing questions	Questioning and challenging		Problem finding	
Acquire the knowledge	Preparation	Exploring data	Learn	Know the domain			Groundwork	Information gathering	
Gather related information			Look		Immersion	Envisaging what might be	Immersion		Observation
Incubation	Incubation	Constructing opportunities	Explore possible strategies	Take time off	Play	Keeping options open		Concept search	
Generate ideas	Insight	Generating ideas		Generate ideas	Being imaginative	Exploring ideas	Divergent exploration	Idea generation	Brainstorming
Combine ideas		Developing solutions		Cross-fertilize ideas		Making connections and seeing relationships		Conceptual combination	
Select the best ideas	Verification			Judging ideas		Reflecting critically on ideas	Selection	Idea evaluation	
Externalize ideas	Elaboration	Building acceptance	Act and anticipate outcomes	Sell the idea, persevere	Self-determination		Articulation of solution, development and transformation, implementation	Implementation planning and action monitoring	Rapid prototyping, refining, implementation

“How can AI help?”

1. **Brainstorming:** using frameworks such as SCAMPER
2. **Inspiring:** finding raw material from new sources
3. **A non-blank page:** first drafts to apply your creativity to
4. **Learning:** new creative skills, processes, strategies
5. **Variations:** to choose from
6. **Realisation:** turn more of your ideas into drafts
7. **Feedback:** different perspectives on your drafts

Strategies

- Prompt design, editing + adapting output as creative acts
- Design prompts that encourage and develop creativity
- Review outputs for originality and diversity
- Identify non-AI creative methods
- Experiment & compare parallel processes (AI/non-AI)
- Develop components of creativity (expertise, creative thinking, motivation)

Key points

- **Compare pros and cons** and make decisions on a case by case basis (with guidelines/checklists to fall back on)
- Take steps to **manage and reduce risks** including prompt design and review
- It's a fast-moving field: **keep up to date**, review regularly and refresh training

Questions?